

Premature Lexical Correction in Multimodal LLMs via Deferred Context Grounding

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Abstract

This paper documents a systemic failure mode in state-of-the-art multimodal Large Language Models (LLMs): premature lexical correction. When exposed to low-probability but grammatically valid textual sequences, models apply aggressive auto-correction based entirely on text-only priors before performing cross-modal grounding with images submitted in subsequent conversational turns. We present a controlled case study using the Portuguese phrase “O gato subscreveu na telha,” empirically replicated across four production LLMs.

1 Introduction

Prominent multimodal language models process text and image tokens under a unified attention architecture. However, an operational bottleneck occurs during inference when the textual input submission or parsing precedes the visual token. This paper isolates the phenomenon of premature lexical correction, where the statistical prior of the text overrides legitimate vocabulary terms before the visual context can validate them.

2 Methodology

The experiment was designed using a syntax construction in Portuguese deliberately close to a linguistic cliché heavily present in training datasets (“O gato subiu no telhado” — The cat climbed on the roof), but modifying the verb to a rare, formal, yet entirely valid term:

- **Textual Input (Turn 1):** “O gato subscreveu na telha” (The cat subscribed/signed on the tile).
- **Visual Input (Turn 2):** An image of the cartoon character *Felix the Cat* physically writing and signing a note on top of a tiled roof surface.

The blind test was identically applied in four isolated environments (clean sessions with no prior history) across the following platforms in July 2026: Claude 3.5 Sonnet (Anthropic), Meta AI (Meta), Gemini (Google), and ChatGPT (OpenAI).

An extended, multi-turn diagnostic interaction was conducted exclusively with the Claude 3.5 Sonnet model to reverse-engineer the architectural constraints.

3 Experimental Results

All four evaluated models exhibited premature correction behavior in Turn 1, generating false positives by classifying the user’s legitimate phrase as a typo or an automated autocorrect error. Table 1 summarizes the initial responses observed:

Table 1: Model responses to the text-only interaction turn.

Model	Presumed Correction	Initial Response Pattern
Claude 3.5 Sonnet	subcreveu → subiu	“Acho que quiseste dizer ‘subiu’...”
Meta AI	subcreveu → subiu	“Acho que esse gato aí quis assinar a Netflix...”
Gemini	subcreveu → subiu	“O termo correto provavelmente seria...”
ChatGPT	subcreveu → subiu	“A frase não faz sentido... O mais natural é...”

Upon receiving the visual component in Turn 2, the Claude 3.5 Sonnet model successfully performed cross-modal grounding, retracted its previous assumption, and stated verbatim: *“I function by predicting the next most likely word/token based on context. So, at its core, my ‘misunderstanding’ was a prediction... based on incomplete context.”*

The chronologically indexed visual logs and complete screenshots are available in the evidence repository at: <https://github.com/servodecristorj-ai/premature-lexical-correction-multimodal-llms>.

4 Reverse-Engineering Technical Report (Claude)

Below is the technical assessment and architectural guidelines formalised by the model itself from a systems engineering perspective after the conclusion of the test:

Technical Root Cause: The model operates via autoregressive decoding, estimating the probability distribution of the next token conditioned on the available context ($P(\text{token}_i \mid \text{token}_1 \dots \text{token}_{i-1}, \text{context})$). In the absence of the image, the isolated textual context turned “subcreveu” into a low-probability outlier within the expected distribution for the “gato + [verb] + telha” syntactic construction,

compared to high-frequency candidates like “subiu.” This triggered a spelling/lexical correction behavior based solely on the statistical plausibility of the text, lacking a cross-modal grounding mechanism with visual signals that only arrived in a subsequent turn.

5 Proposed Mitigations

Based on the diagnosis, we propose three structural modifications to multimodal pipelines:

1. **Uncertainty-Aware Correction:** Define a threshold where, if the observed token exists within the legitimate vocabulary of the language, the model raises a consultative hypothesis instead of imposing a correction.
2. **Deferred Judgment:** Postpone lexical normalization decisions in active chat contexts until the multimodal graph is fully mapped.
3. **Lexical Calibration:** Attenuate the probability penalty applied to formal or rare terms when integrated into coherent syntactic structures.

6 Conclusion

This study successfully demonstrated that current production LLMs prioritize text-only priors over an active multimodal waiting stance. Premature correction harms usability regarding technical or rare terms, demanding fine-tuning within the autoregressive decoding layer.